IoT Connectivity Technologies

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Fall Semester

IoT Connectivity Technologies

Learning objectives

- Identify key connectivity requirements
- Explain wireless communication basic
- Compare major IoT connectivity technologies
- Explain common network topologies and gateways
- Select suitable connectivity for an IoT application

Application requirements

↓

Connectivity requirements

↓

Physical communication constraints

↓

Candidate connectivity technologies

↓

Technology comparison

↓

Connectivity selection

Contents

1. IoT Connectivity Fundamentals
2. Local and Short-Range Connectivity Technologies
3. Wide-Area Connectivity and Technology Selection

1. IoT Connectivity Fundamentals

Connection From Week 3

- Week 3 focuses on what IoT device sends

Physical quantity

↓

Sensor

↓

IoT Device

↓

Processing

↓

Structured data

↓

Serialized format

- Week 4 focuses on how device establishes a link to send

Serialized format

↓

Communication interface

↓

Connectivity technology

↓

Receiver / Gateway / Network

1. IoT Connectivity Fundamentals

What is IoT connectivity?

IoT connectivity

- Wired or wireless communication link
  - ✓ Allows an IoT device to **exchange data**
    - With another **device**, **gateway**, **access point**, **base station**, or **network**

Examples:

- Wearable – Bluetooth → Smartphone
- Camera – Wi-Fi→ Access Point
- Machine – Ethernet→ Industrial Network
- Farm Sensor – LoRa→ Gateway
- Tracker – Cellular→ Base Station

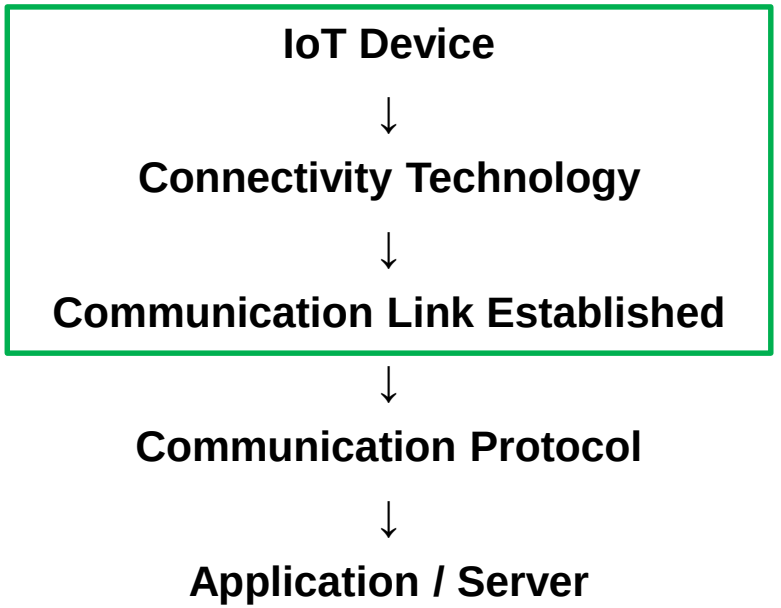
Different connectivity paths carry IoT data

How is the device connected?	Wi-Fi, BLE, Ethernet, LoRa, cellular
How is data later communicated over the network?	Communication protocol → Week 5

1. IoT Connectivity Fundamentals

Connectivity vs. Communication protocol

- **Connectivity** – how the devices get connected
- **Communication protocols** – how data is exchanged once connection exists



1. IoT Connectivity Fundamentals

Connectivity vs. Communication protocol

- **Assume a sensor uses Wi-Fi to connect with nearby network**

  → Wi-Fi provides wireless communication link

- But, Wi-Fi does not define rules for IoT application exchanges sensor messages with a server

  → Involve communication protocol once connection is established

Connectivity technology = Road / highway

Communication protocol = Rules for how information travels on that road

Device

↓

Wi-Fi → A wireless link exists

↓

Network

To answer:

- **Where** is the destination?
- **How** should the **data be delivered**?
- **What happens** if communication **fails**?
- **How** should an application **request or publish information**?

→ **Protocol layers**

1. IoT Connectivity Fundamentals

Why not use Wi-Fi for everything?

❖ Consider different applications with different characteristics

Application	Main characteristics
Security camera	High data volume, fixed, power available
Wearable health sensor	Small data volume, short range, battery
Farm temperature sensor	Small data volume, long range, battery
Delivery vehicle tracker	Wide area, moving, periodic data

Could one connectivity technology satisfy all four applications efficiently?

Technically possible ≠ architecturally appropriate

1. IoT Connectivity Fundamentals

Why not use Wi-Fi for everything?

1. Security camera

Characteristics:

- Large amount of data
- Continuous video
- Fixed location
- Power usually available
- Local network available

A camera may need to send:

- Video frames
- Audio
- Status information

→ **High data rate**

Suitable for use:

Ethernet, Wi-Fi

2. Wearable health sensor

Characteristics:

- Small amount of data
- Battery powered
- Device worn on body
- Phone usually nearby

Main constraints:

- Very low power
- Short/local distance
- Mobility
- small data volume

Device – [BLE] – smartphone

Suitable for use:

BLE (Bluetooth Low Energy)

1. IoT Connectivity Fundamentals

Why not use Wi-Fi for everything?

3. Farm temperature sensor

Characteristics:

- Low data rate
- **Long range**
- Battery or solar powered
- Very small payload

Data might be only: ~ 10 bytes

```
{
    "temperature": 24.3,
    "battery": 81
}
```

**** Distance ****

Suitable for use: LoRa (Long range)

4. Delivery vehicle tracker – GPS, battery, cargo temperature

Characteristics:

- Moving
- City-wide / regional operation
- Moderate or low data
- No fixed nearby gateway

Main requirements:

- Mobility
- Wide-area coverage

Vehicle – [cellular] – base station

Suitable for use: LTE, cellular

1. IoT Connectivity Fundamentals

Why not use Wi-Fi for everything?

❖ **To decide which technology to use:**

First ask: “What does this application **need**?”

Then: “Which **technology matches** those requirements?”

Application requirements

↓

Connectivity requirements

↓

Candidate technologies

↓

Best-fit technology

Camera: high data rate → Ethernet / wiFi

Wearable device: battery → BLE

Farm sensor: long local range → LoRa

Vehicle tracker: continuous mobility over large area → LTE / Cellular

Common connectivity requirements:

- Range
- Data rate
- Power
- Mobility

1. IoT Connectivity Fundamentals

Connectivity requirement 1: Range

- **Range**
  - Maximum practical communication distance between communicating nodes
- **Scale:**

Centimeters

↓

Meters

↓

Tens/hundreds of meters

↓

Kilometers

↓

Regional/global

- **Example:**
  - NFC (near-field communication): centimeters 
    - Contactless payment, transit passes
  - BLE: local/short
  - Wi-Fi: local area
  - LoRa: kilometers
  - Cellular: wide area
  - Satellite: extremely wide area

1. IoT Connectivity Fundamentals

Connectivity requirement 1: Range

“Range = 100 m”

does NOT mean

“always works at 100 m”

❖ **Real range depends on:**

- Transmitter power
- Antenna
- Frequency
- Receiver sensitivity
- Obstacles
- Interference
- Environment

100 m open field ≠ 100 m through factory walls

1. IoT Connectivity Fundamentals

Connectivity requirement 2: Data rate

- **Data rate**
  - **Amount of digital data** transferred per unit time, typically bit/s

Temperature sensor

25–100 bytes

every 1–10 minutes

Temperature sensor does not need 1 Gbit/s link

Video camera

Continuous stream

Possibly millions of bits per second

1 Gbit/s link

- transfers data at rate of 1 billion bits per second
  ≈ 125MB/s

1. IoT Connectivity Fundamentals

Connectivity requirement 2: Data rate

Bandwidth vs. Data rate

- **Bandwidth**
  - Maximum amount of data that can be transmitted over an internet or network connection in a given amount of time
- **Data rate**
  - Rate at which digital information is transferred

Greater available bandwidth can support **higher data rates,**

but the two quantities **are not identical**

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

- **Wireless transceiver (radio)** can operate in **different states** such as:
  - Transmit
  - Receive
  - Idle/listen
  - Sleep
- **Communication energy is determined by radio behavior over time**
  - IoT radio does not consume the same amount of power all the time
  - Consumption depends on its operating state
- Each state has different power demand
  - Transmit / Receive → high activity
  - Idle / Listening → moderate activity
  - Sleep → very low activity

For battery-operated IoT devices, design objective:

- Keep the radio asleep as much as possible
- Activate it only when communication is necessary

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

Example 1: Mains-powered security camera

- A **security camera** may have:
  - Continuous electrical power
  - Continuous wi-fi or ethernet connectivity
  - High-volume video transmission
  - Frequent communication

- For this type of device:

Power available continuously

↓

Radio can stay active longer

↓

Higher communication energy may be acceptable

Hence, communication energy is NOT dominant constraint

Instead, **prioritize**:

- ✓ data rate
- ✓ stable connection
- ✓ Latency
- ✓ reliability

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

Example 2: Coin-cell wearable

- A **wearable** may have:
  - Very small battery
  - No convenient charging
  - Expected operating lifetime of months
  - Very small sensor data
- Example:

  Heart rate = 78 bpm

  Temperature = 36.7°C → **data is small**

  Battery = 83%

- If the wearable **keeps its radio active continuously**,
  - Battery may drain quickly even though actual amount of useful information is small
- Instead, **the device may operate in the way**:

Sleep for most of the time

↓

Wake briefly

↓

Measure sensor

↓

Transmit a small message

↓

Return to sleep

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

Why radio activity matters?

$$battery\ consumption \propto I \times t$$

where: I = current drawn by radio (mA); t = active time (hr)

Assume two devices radio draw: 30 mA

DEVICE A

Transmit 1 second every minute

Over 24 hours:

60 × 24 = 1440 transmission/day

Transmission time:

→ 1440 × 1 = 1440 seconds/day = 0.4 h/day

Daily transmit consumption:

30 × 0.4 = 12 mAh/day

DEVICE B

Transmit 1 second every hour

Over 24 hours:

24 transmission/day

Transmission time:

→ 24 seconds/day = 0.00667 h/day

Daily transmit consumption:

30 × 0.00667 = 0.20 mAh/day

Device A consumes ≈ 60 times transmit energy

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

Real battery consumption is more complicated

- **A real IoT device consumes energy during:**
  - Sensor measurement
  - Processor operation
  - Radio startup
  - Transmission
  - Reception
  - Listening
  - Waiting
  - Retransmission
  - Sleep

Total battery consumption is more like:

$$E_{total} = E_{sensor} + E_{processing} + E_{transmit} + E_{receive} + E_{idle} + E_{sleep}$$

→ **Communication frequency and radio-on time strongly influence battery life**

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

Duty cycle

❖ **Fraction of time a device or radio is active compared with the total time**

- **Example:**
  - ***Device A***: Radio active 1 second every 60 seconds (1 minute)
    - **Approximate duty cycle**: $\frac{1}{60} \times 100 \approx 1.67\%$
  - ***Device B***: Radio active 1 seconds every 3600 seconds (1 hour)
    - **Approximate duty cycle**: $\frac{1}{360} \times 100 \approx 0.028\%$

➢ **Lower duty cycle** means:

Less radio-on time

↓

Lower communication energy

↓

Longer battery life

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

- **More data can mean more radio-on time**

- **Example:**
  - ***Device A***: Payload = 20 bytes
  - ***Device B***: Payload = 2MB

- **More data** means:

More transmission time

↓

More radio activity

↓

More energy

1. IoT Connectivity Fundamentals

Connectivity requirement 3: Power consumption

“If a temperature sensor changes very slowly, should it transmit every second?”

Probably not.

The application may not need:

- Frequent update rate
- Unnecessary communication consumes:
  - Battery
  - Network resources

“If it is an emergency gas-leak sensor?”

For the gas sensor, needs:

- Low latency (Minimal delay)
- Rapid alarm

Connectivity design always
balances multiple requirements

1. IoT Connectivity Fundamentals

Connectivity requirement 4: Latency

- **Latency**
  - **Delay** between **sending** information and its **arrival/use at the destination**

- **Example:**

Application	**Delay tolerance**
Soil moisture monitoring	seconds/minutes may be acceptable
Building temperature	seconds often acceptable
Emergency alarm	much lower delay desired
Time-sensitive control	Very low delay may be required

1. IoT Connectivity Fundamentals

Connectivity requirement 5: Mobility

❖ **In IoT connectivity, mobility describes**

- ➢ Whether a device changes its physical location while the system is operating
- ➢ Whether communication must continue while it moves

➢ **Fixed device**:

- Once installed, communication is stable
- Example:
  - Factory machine
  - Temperature sensor

➢ **Mobile device**:

- Device changes location while operating
- Example:
  - Smartwatch
  - Delivery vehicle
  - Mobile robot

❖ **Different levels of mobility**

Mobility level	Example	Communication requirement
Fixed	Factory machine	Device stays in one place
Local mobility	Smartwatch	Moves within a limited local area
Wide-area mobility	Scooter / vehicle	Moves over large geographic areas

1. IoT Connectivity Fundamentals

Connectivity requirement 5: Mobility

➢ **Example – Factory machine**:

- Characteristics:
  - Location does not change
  - Cable installation is possible
  - Continuous connectivity may be required
  - **Mobility is irrelevant**

Factory Machine

↓

Ethernet

↓

Industrial network

No mobility required → Ethernet

➢ **Example – Smartwatch**:

- Smartwatch can move:
  - While walking
  - While exercising
  - Throughout normal daily activity

→ **Local mobility**

→ **BLE** supports mobility **within communication range of the phone**

1. IoT Connectivity Fundamentals

Connectivity requirement 5: Mobility

➢ **Example – Rental scooter**:

- Requirements:
  - Moves throughout a city
  - Needs location reporting
  - Needs battery/status reporting

A scooter may travel:

Area A

↓

Area B

↓

Area C

↓

Area D

Communication system needs to remain available across large geographical area → **Wide-area mobility**

Scooter

↓

Cellular

↓

Cellular infrastructure

WiFi less suitable:

→ Once the scooter leaves coverage area, no WiFi connection

1. IoT Connectivity Fundamentals

Connectivity requirement 5: Mobility

- **Range vs. Mobility**

	Range	Mobility
Main question	How far can the link reach?	Can communication continue as the device moves?
Example	Sensor 3 km from gateway	Scooter travelling through city
Relevant factor	Radio/link distance	Coverage + infrastructure while moving

- **Mobility requirement:**

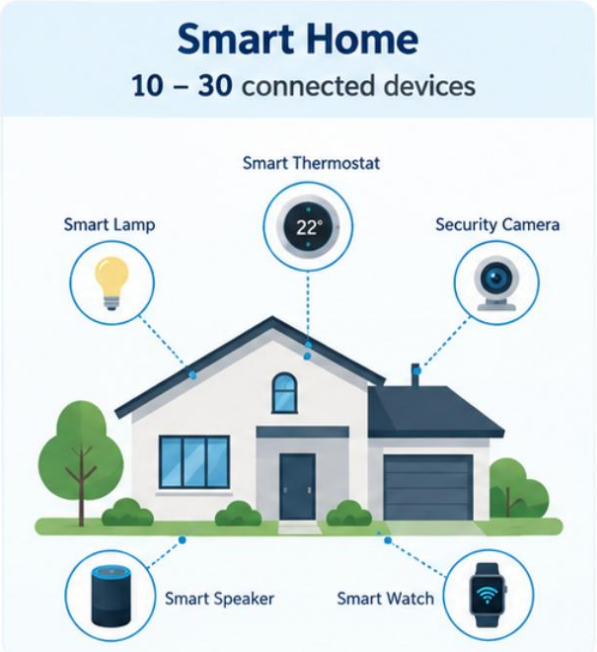
  How far can the device move while still needing communication?

1. IoT Connectivity Fundamentals

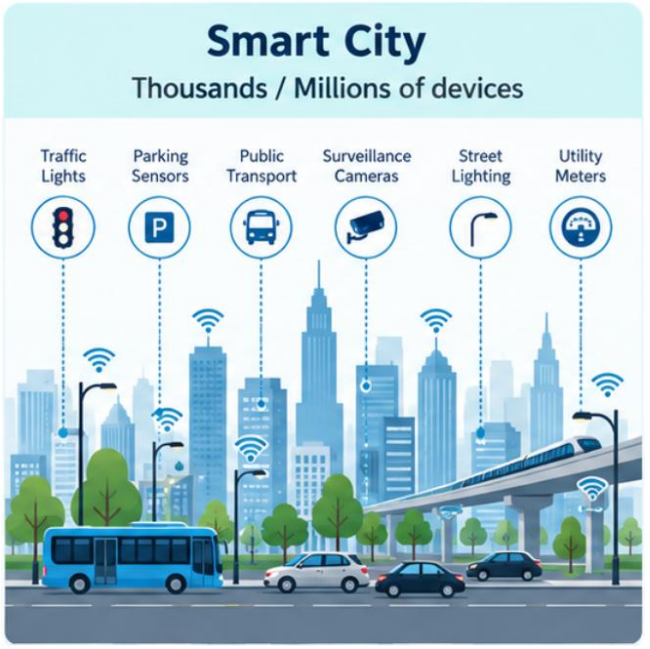
Device density

Connectivity design must consider:

- Not only one device
- But how many devices compete or share communication infrastructure







Increasing device scale

1. IoT Connectivity Fundamentals

Infrastructure

- **IoT technology** selection depends on what communication infrastructure is available



- **Example 1: Wearable → smartphone**

A wearable connects to smartphone using Bluetooth which is already available



- **Example 2: Farm sensor → no WiFi nearby**

A farm sensor – remote field with no WiFi access point nearby

→ Another technology is needed

→ E.g., cellular

1. IoT Connectivity Fundamentals

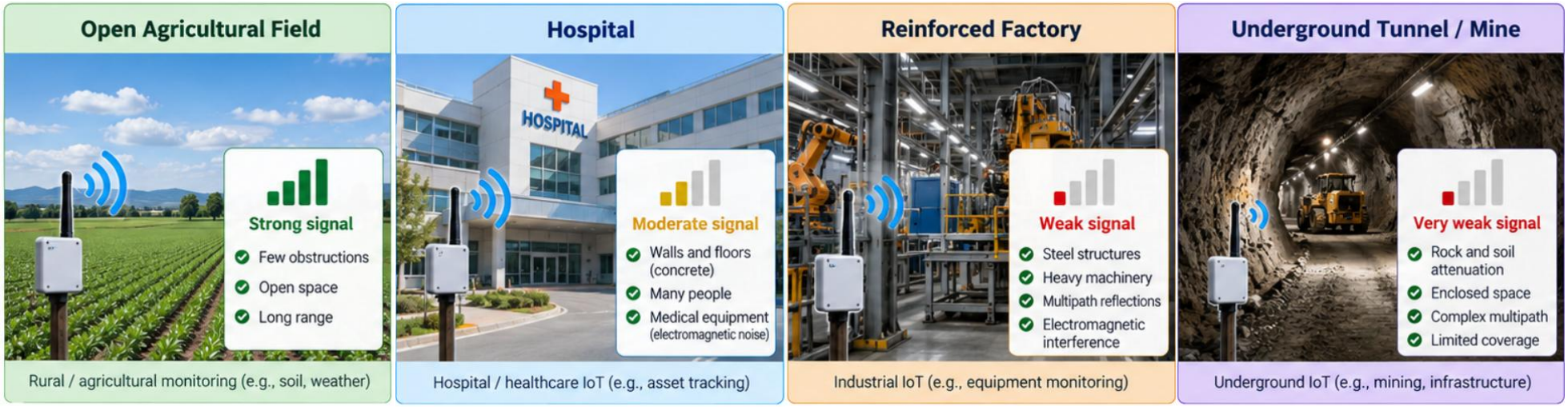
Environment

❖ Same IoT device can perform differently depending on physical environment



➢ **Key environmental factors affect wireless performance:**

- Concrete – weaken signals, absorbs radio energy
- Steel – reflects and blocks signals
- Machinery – generates electromagnetic interference
- Underground structures – severe weakening, blocks signals



1. IoT Connectivity Fundamentals

Cost and reliability

❖ **Cost can include:**



❖ **Reliability questions**

- Is occasional loss acceptable?
- Is connection availability critical?
- Is infrastructure controlled privately?
- Is redundant coverage (backup) required?

Connectivity selection depends on:

➢ **Engineering performance**

➢ **Deployment constraints**

1. IoT Connectivity Fundamentals

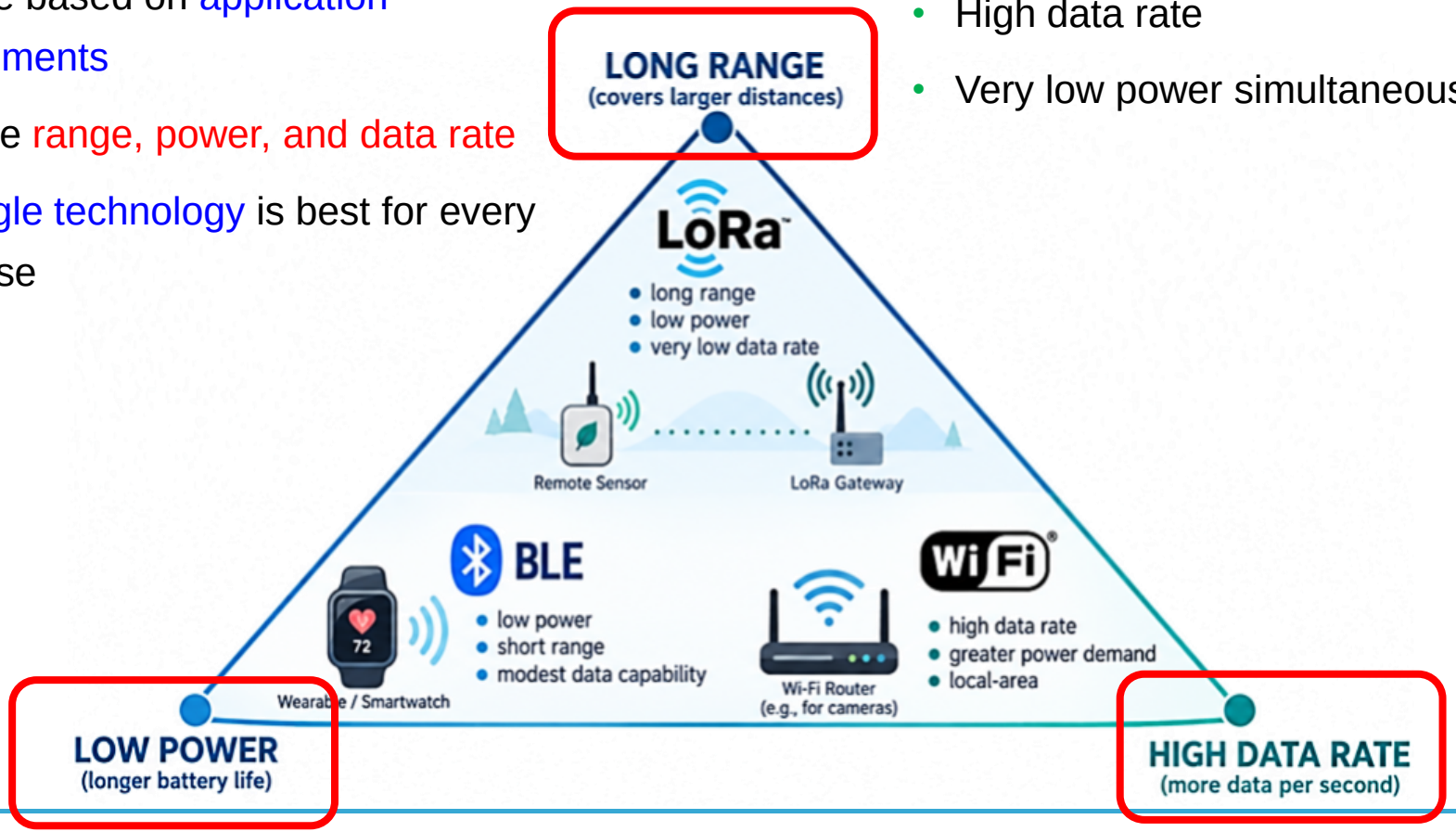
Connectivity trade-off triangle

❖ **Different technologies, different strengths**

- Choose based on application requirements
- Balance range, power, and data rate
- No single technology is best for every use case

❖ **A system rarely maximizes:**

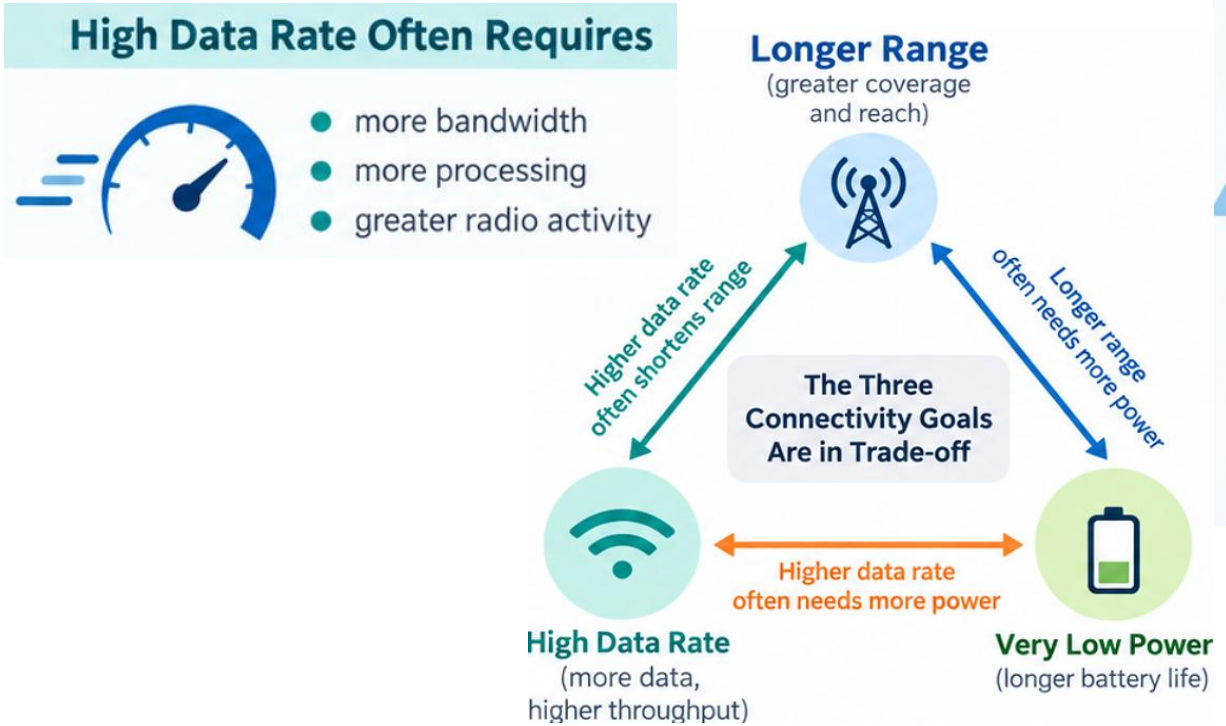
- Long range
- High data rate
- Very low power simultaneously

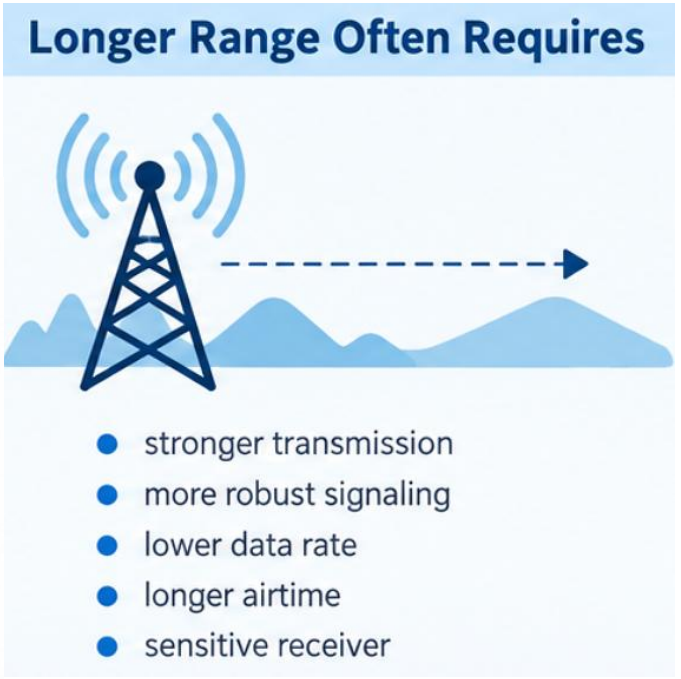


1. IoT Connectivity Fundamentals

Connectivity trade-off triangle

❖ **Reasons why cannot maximize all three**







1. IoT Connectivity Fundamentals

Short discussion

Scenario A – Security camera

- Fixed
- Continuous power
- High data
- Local building

Ethernet / WiFi

Scenario B – Wearable heart-rate patch

- Battery
- Small data
- Nearby phone
- Mobile person

BLE

Scenario	Range requirement	Data requirement	Power constraint	Mobility	Priority requirement
A. Security camera	Local building	**High –** continuous video	Low concern if mains-powered	Fixed	Data rate
B. Wearable heart-rate patch	Short range	**Low** – small periodic measurements	**Very important** battery **–** powered	Moves with person	Low power

1. IoT Connectivity Fundamentals

Short discussion

Scenario C – Farm soil sensor

- Fixed
- Several kilometers
- Tiny message
- Battery

LoRa

Scenario D – Delivery tracker

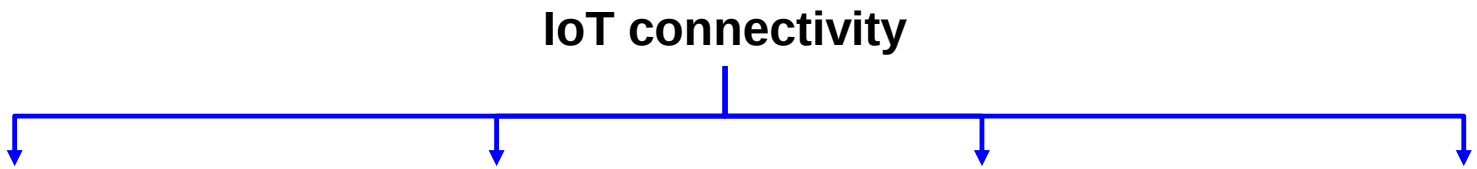
- City-wide
- Moving
- Battery
- Periodic status

Cellular

Scenario	Range requirement	Data requirement	Power constraint	Mobility	Priority requirement
C. Farm soil sensor	Long range – several km	**Very low –** tiny periodic message	**Very important** battery **–** powered	Fixed	Long range, low power
D. Delivery tracker	Wide area – city/regional	**Low to moderate** – periodic location	**Important**	Highly mobile	Mobile, wide-area coverage

2. Local and Short-Range Connectivity Technologies

Connectivity categories



Wired	Local / short range wireless	Identification / very short range	Wide area
Ethernet	Wi-Fi	RFID	LoRa
	BLE	NFC	NB-IoT
	IEEE 802.15.4		LTE-M
			4G/5G
			Satelite

2. Local and Short-Range Connectivity Technologies

Wired vs. wireless

Wired	Wireless
Physical cable	Radio
Stable link	Flexible deployment
High throughput possible	Mobility possible
Less radio frequency interference	Subject to propagation/interference
Installation needed	Coverage constraints

2. Local and Short-Range Connectivity Technologies

Wired – Ethernet

- **Strengths**
  - Very high data rate
  - Stable
  - Reliable physical link
  - Low susceptibility to RF interference
  - Suitable for continuous communication
- **Limitations**
  - Physical cable required
  - Installation effort
  - Limited mobility
- **Typical IoT roles**
  - Industrial machines
  - Gateways
  - Fixed cameras
  - Building equipment

PoE

- Allows electrical power & network data to be delivered
  - Over the Ethernet cabling infrastructure
- Useful for:
  - IP camera
  - Wireless access point
  - Gateway
  - Industrial sensor/controller

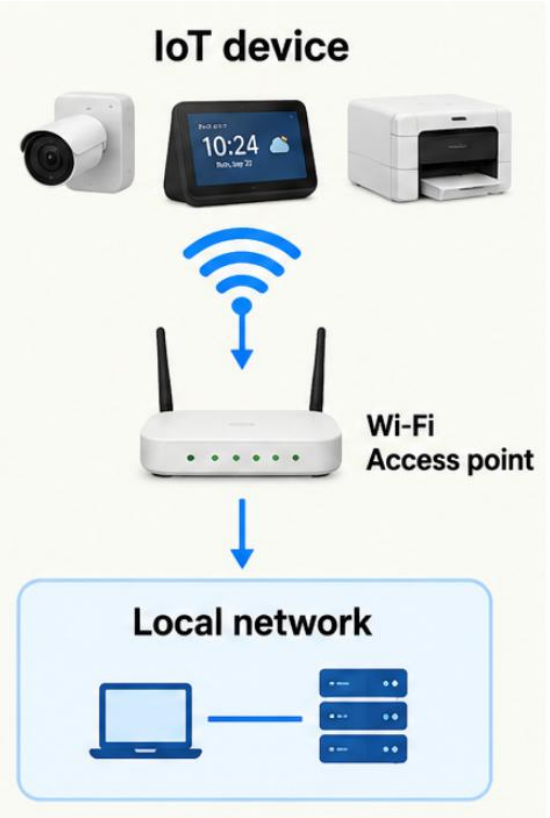
- **One cable → data + power**

2. Local and Short-Range Connectivity Technologies

Wi-Fi

❖ **Characteristics**:

- Local-area
- High throughput
- Existing infrastructure common
- Relatively high power compared with low-power IoT radios



2. Local and Short-Range Connectivity Technologies

Wi-Fi

❖ **Best for local-area devices with existing Wi-Fi infrastructure**





2. Local and Short-Range Connectivity Technologies

Wi-Fi

❖ Limitations of Wi-Fi





- Limited range, especially in rural areas



Requires:

- Access point
- Power

Which may not available or economical to deploy



- **Wi-Fi** – for **high data rate**
- Sensor only sends very small amount of data



- **Wi-Fi** consumes relatively high power

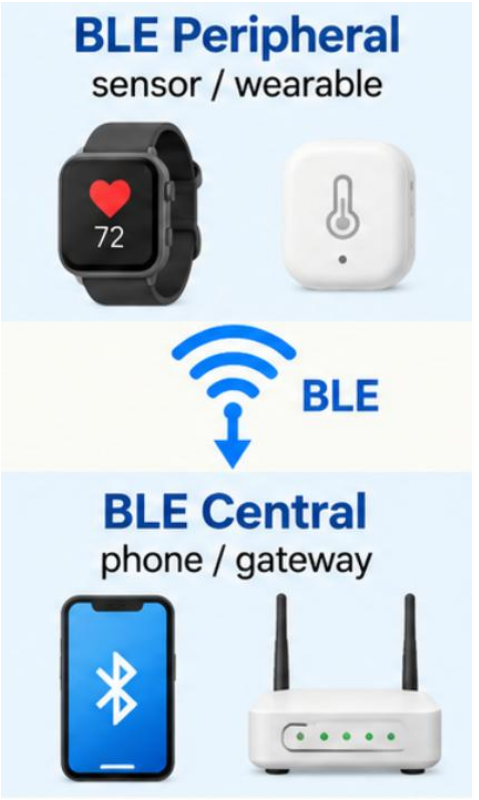
→ Drains coin-cell battery

2. Local and Short-Range Connectivity Technologies

BLE – Bluetooth Low Energy

Bluetooth Low Energy (BLE)

- **Low power communication** designed for **battery-powered devices**
- **Short-range** (tens of meters) for personal and nearby environments
- Commonly used for **device-to-phone** or device-to-gateway communication
- Ideal for **sensor data** and **simple control** with **low data rates**

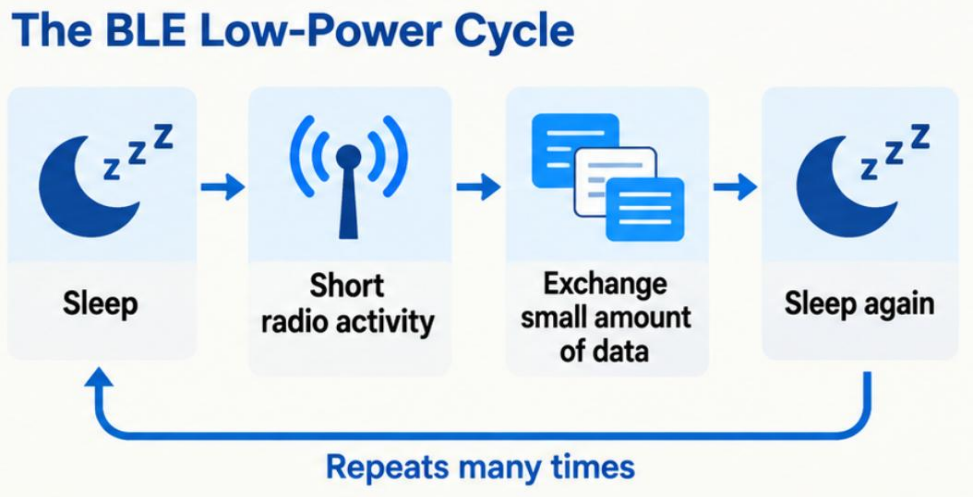


2. Local and Short-Range Connectivity Technologies

BLE – Bluetooth Low Energy

Why BLE is low power

- **Radio stays off** most of time
- BLE **wakes briefly** to send or receive small data
- **Short active time** reduces energy use
  - → Supports **battery-powered devices**
  - → Saves power by keeping **radio active** only for **short period**



2. Local and Short-Range Connectivity Technologies

BLE – Bluetooth Low Energy

Examples of BLE use cases



Syncs data with phone (notifications, health data)



Continuously monitor vital signs (health rate, temperature



Tracks activity, steps and sleep



Use for indoor navigation, asset tracking



Sends sensor readings to smartphone (temperature, humidity, environmental data)

BLE fits for:



2. Local and Short-Range Connectivity Technologies

Wi-Fi vs. BLE

Requirement	Wi-Fi	BLE
Data rate	Higher	Lower
Power	Higher	Very low
Typical range	Local	Short/local
Typical infrastructure	AP	Phone/gateway
Typical application	Camera	Wearable

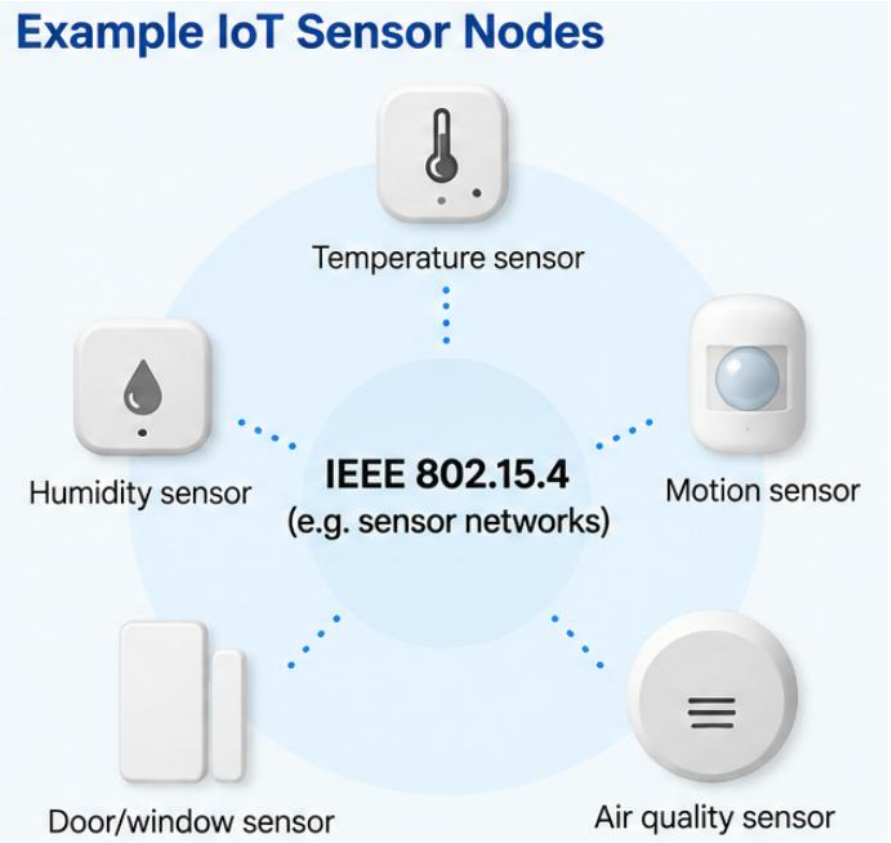
2. Local and Short-Range Connectivity Technologies

IEEE 802.15.4

- **IEEE 802.15.4**
- Specifies a low-rate, low power wireless foundation used by several IoT networking technologies

Key characteristics:

- **Low data rate**
  - Designed for small, infrequent data transmission (e.g., sensor measurement)
- **Low power**
  - Suitable for battery-powered devices
- **Short to medium range**
  - Tens to hundreds of meters
- **Suitable for many sensor nodes**
  - Well-suited for IoT applications with many low-power devices

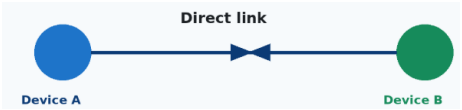


2. Local and Short-Range Connectivity Technologies

Network topology

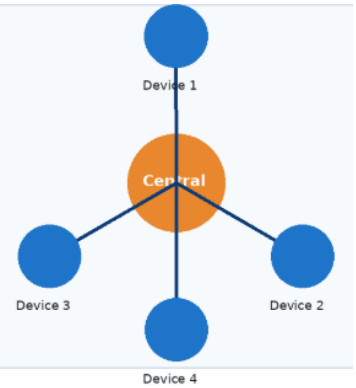
❖ **Topology describes how communicating nodes are arranged and connected**

Point-to-point topology



Two endpoints communicate directly

Star topology



Multiple devices connect through one central node

Advantages:

- Simple
- Easy centralized management

Limitation:

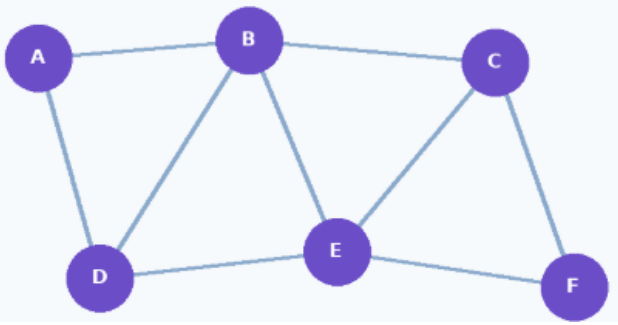
- Central point dependency
- Every endpoint must reach central

2. Local and Short-Range Connectivity Technologies

Network topology

❖ **Topology describes how communicating nodes are arranged and connected**

Mesh topology



Nodes can connect through multiple neighboring nodes

Advantages:

- Extended coverage
- Alternative paths

Trade-offs:

- More complexity – maintain relationships among nodes
- Relay burden
  - One node needs to send own data and relay another node's data

More relay activity

↓

More radio-on time

↓

More energy consumption

2. Local and Short-Range Connectivity Technologies

Network topology

Topology	Communication arrangement	Simple example
Point-to-point	One device directly connected to another	Sensor ↔ controller
Star	Many devices connect to one central node	Devices → Wi-Fi AP
Mesh	Devices may connect through neighboring devices	Building sensor network

2. Local and Short-Range Connectivity Technologies

RFID – Radio Frequency Identification

RFID

- **Wireless technology** that uses radio waves
  - To automatically identify and track tags attached to:
    - Objects, animals, or people without requiring a direct line of sight
  - Line of sight (LoS) – unobstructed straight path through air between transmitting antenna and receiving antenna



RFID Tag
Stores a unique identifier
(Electronic product code)



RFID Reader
Sends RF signals, powers the tag (for passive tags), and reads the data.



Digital System
Processes the data and integrates it with business applications (e.g., database, ERP).

Key applications:



Identification
Uniquely identify objects, people or assets.



Tracking
Monitor location and movement in real time.



Inventory
Automate and streamline inventory management.

2. Local and Short-Range Connectivity Technologies

RFID – Radio Frequency Identification

IoT use of RFID: Identifying an object

Warehouse pallet



Scan RFID tags on pallets or boxes to quickly identify and track inventory.

Package identification



Read an RFID tag on a parcel to identify it throughout the supply chain.

Asset tracking



Identify and locate equipment and other assets with RFID tags.

Access control



Use RFID badges or cards to identify people and grant access to secure areas.

2. Local and Short-Range Connectivity Technologies

RFID – Radio Frequency Identification

Feature	Barcode	QR Code	RFID
Reading method	Optical	Optical	Radio wave
Line of sight	Required	Required	Not required
Multiple items	Usually one at a time	Usually one at a time	Multiple tags possible
Cost	Very low	Very low	Higher
Typical use	Product ID	Links/tickets/info	Inventory/assets







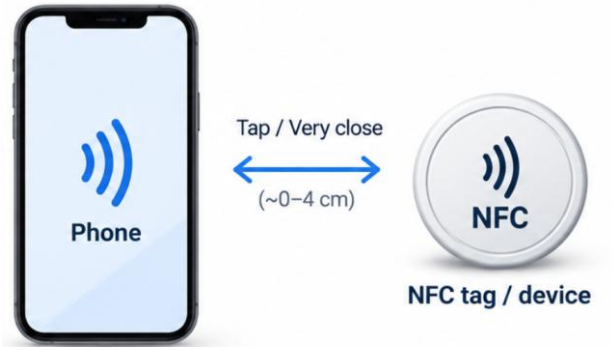
2. Local and Short-Range Connectivity Technologies

NFC – Near Field Communication

Very short range
Typically a few centimetres (~0–4 cm).

Deliberate proximity
Requires intentional close placement (tap).

Identification / Pairing / Access / Payment
Used for secure, short-range data exchange.



2. Local and Short-Range Connectivity Technologies

NFC – Near Field Communication

Smartphone tap-to-pay



Tap your smartphone at a payment terminal to make a secure, quick payment.

Contactless payment card



Tap a contactless payment card at a terminal to pay for goods and services.

Transport card



Tap a transport card at a subway or bus fare gate to enter quickly.

Camera-to-smartphone pairing



Tap or hold your smartphone close to an NFC-enabled camera to pair quickly and transfer images.

2. Local and Short-Range Connectivity Technologies

Local technology comparison

Technology	Relative range	Data rate	Power	Typical use
Ethernet	cable	very high	usually powered	machine/gateway /camera
Wi-Fi	local	high	medium–high	camera/appliance
BLE	short	low–medium	very low	wearable
802.15.4/Zigbee	short–medium	low	low	sensor network
RFID/NFC	very short/reader-dependent	low	very low/passive	identification

2. Local and Short-Range Connectivity Technologies

Short discussion

Based on:

- Range reason
- Data-rate reason
- Power reason
- Infrastructure reason

Decide communication technology for:

1. Smartwatch → phone
2. Machine-vision camera → control room
3. Warehouse pallet identity
4. Building temperature sensors

2. Local and Short-Range Connectivity Technologies

Short discussion

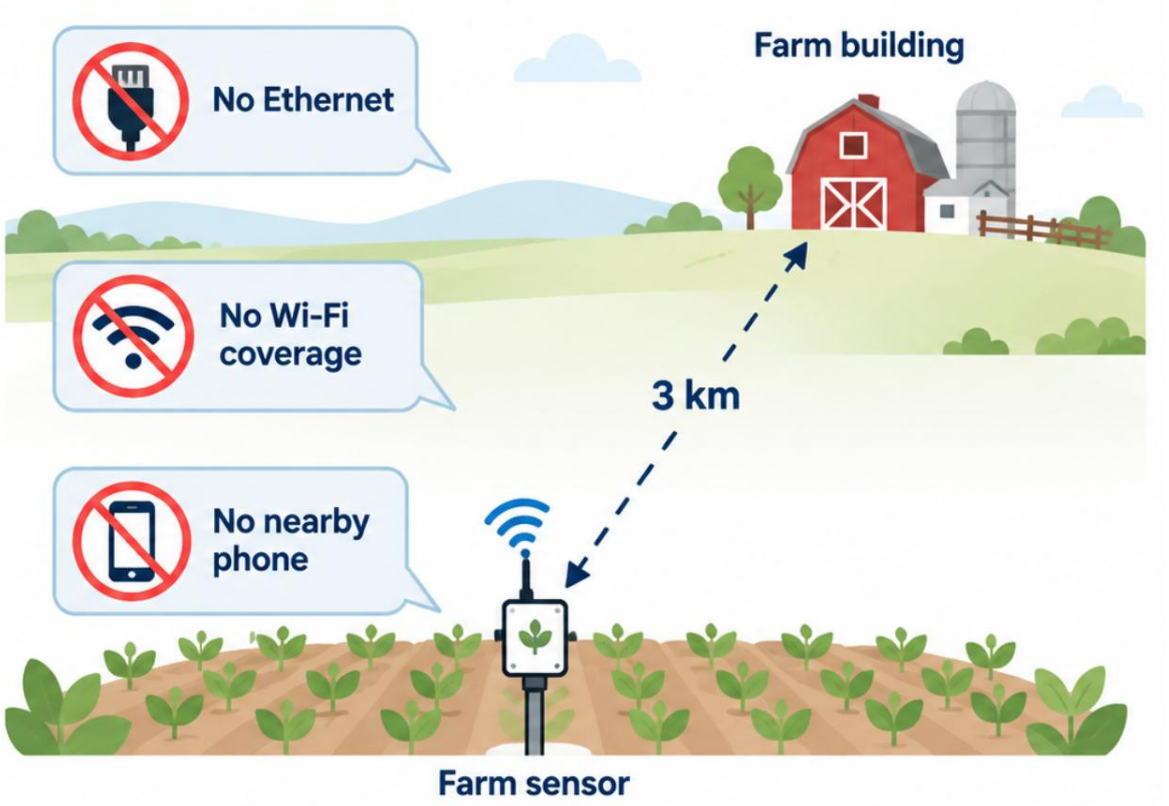
Case	Technology	Range reason	Data-rate reason	Power reason	Infrastructure reason
Smartwatch → phone	**BLE**	Short-range communication	Small/moderate	Very low power	Smartphone already supports BLE and acts as the nearby central device
Machine-vision camera → control room	**Ethernet or Wi-Fi**	Local scale communication	Video requires high data rate	Camera is usually mains-powered	Ethernet is preferred if cabling exists; Wi-Fi is useful if cabling is difficult and an AP is available
Warehouse pallet identity	**RFID**	Only reader-to-tag range	Only a small identifier needs to be transferred	Passive RFID tags may operate without battery	Install fixed or handheld RFID readers
Building temperature sensors	**IEEE 802.15.4**	Short-to-medium range; multiple nodes can cover a building	Temperature readings → very small and periodic	Low power	Coordinator can collect data from many sensor nodes

3. Wide-Area Connectivity and Technology Selection

Wide-area connectivity

❖ **Why local connectivity is not enough**

- In rural and remote areas, local connectivity may not be available





3. Wide-Area Connectivity and Technology Selection

Local vs. wide-area connectivity

Local connectivity

Short range connection

Provides local access to nearby devices or local network



Wide-area connectivity

Broader-area connectivity

Connects devices over long distance, across cities, regions or globally



Application examples:



3. Wide-Area Connectivity and Technology Selection

LPWAN – Low-power-wide-area network

❖ **LPWAN – wireless technologies designed for IoT that:**

- Provides long range communication with very low power consumption



Trade-off:

Very low data rate

Ideal use case:



Poor fit:



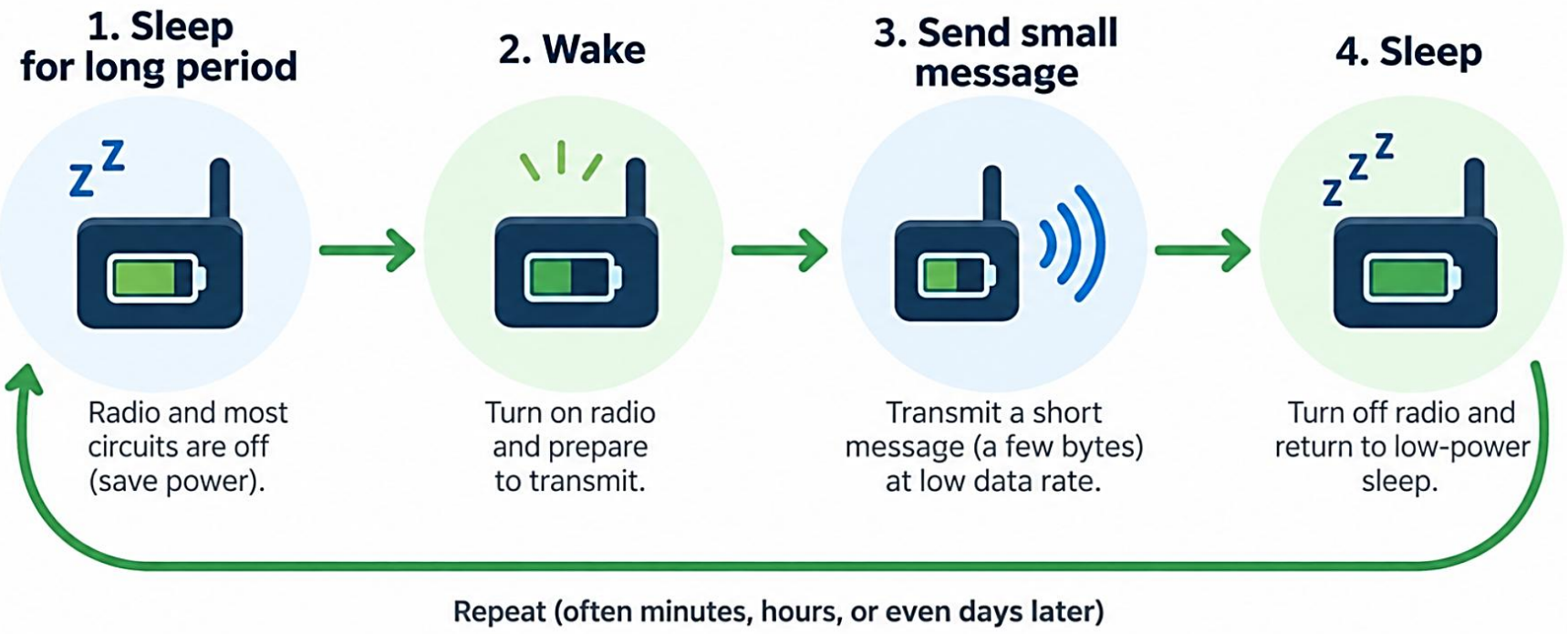
Main LPWAN categories:

Cellular LPWAN	**Non-cellular LPWAN**
(license spectrum)	(Unlicensed spectrum)
Run by telecom operators	Run on open radio bands
Example: NB-IoT, LTE-M	Example: LoRa

3. Wide-Area Connectivity and Technology Selection

Why LPWAN can use little power

❖ **Battery benefits from low duty cycle**



3. Wide-Area Connectivity and Technology Selection

LoRa

- Long-range, low power radio modulation / physical layer technology
- Shaping radio waves → travel long distances on very low power with low data rates
- **Key characteristics:**

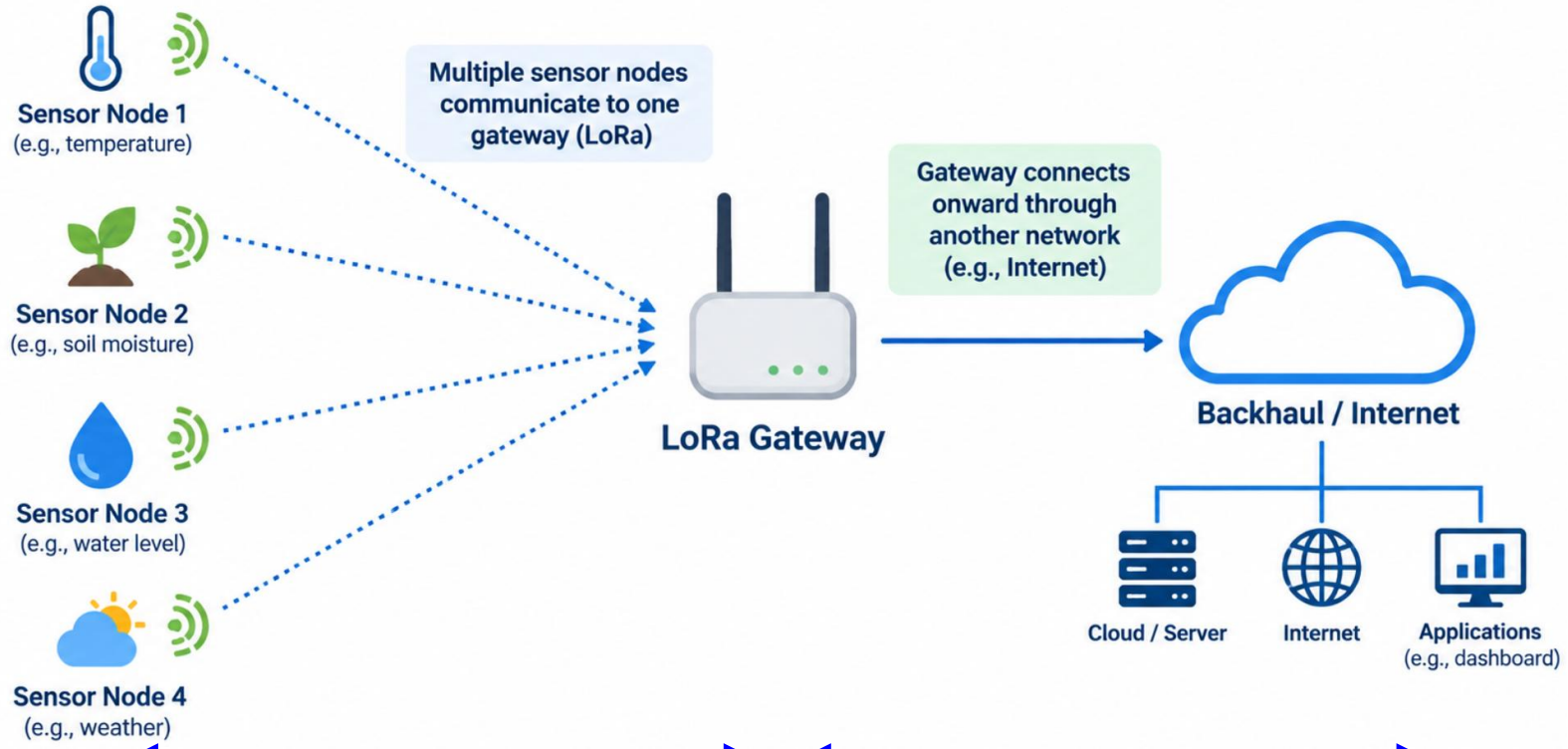




3. Wide-Area Connectivity and Technology Selection

LoRa architecture

- In real deployments, **LoRa is used with LoRaWAN** (network protocol) to build complete IoT systems



3. Wide-Area Connectivity and Technology Selection

LoRa vs LoRaWAN

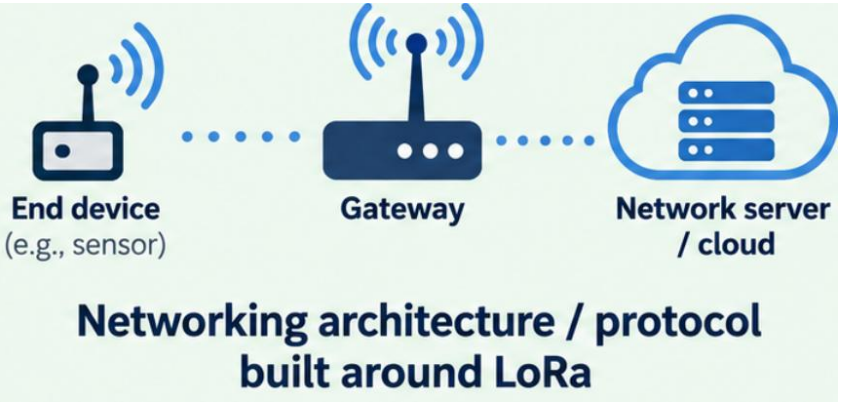
LoRa

Long range, **low power** radio modulation used to transmit data over the air



LoRaWAN

Defines how devices, gateways and network services **communicate**



3. Wide-Area Connectivity and Technology Selection

LoRa strengths and limitations

Strengths



Long range



Low power



Private gateway possible



Useful in remote sensing

Limitations



Very low data rate



Not for high-bandwidth applications

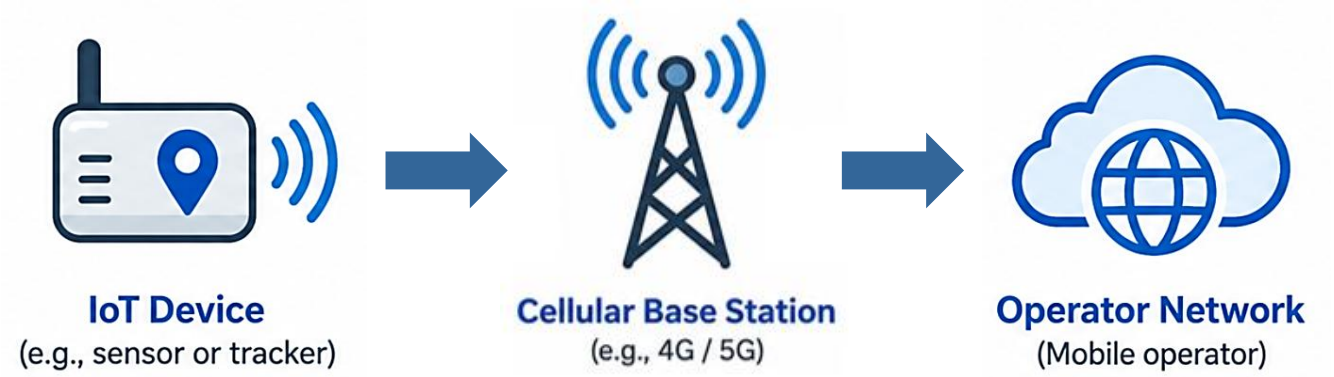


Gateway / network infrastructure still required

3. Wide-Area Connectivity and Technology Selection

Cellular IoT concept

- **IoT devices** use cellular networks to connect to the internet via a mobile operator
- Key characteristics:
  - Wide geographic coverage
  - Operator-managed infrastructure
  - Mobility support
  - Subscription / service dependency



Collects data and sends it over cellular network

Transmits device data to the operator network

Routes data to internet and services (e.g., IoT platform

3. Wide-Area Connectivity and Technology Selection

LoRa vs Cellular

	LoRa	Cellular
Primary Target	Simple machines & industrial sensors	Smartphones, tablets, and laptops
What it transmits	Tiny numbers (e.g., GPS points, temperature)	High-definition video, websites, apps, voice
Bandwidth	Extremely low (A few bytes per message)	Massive (Gigabytes per second)
Network Infrastructure	Private or Public	Public Only (Must use a mobile carrier's cell tower)
Battery Consumption	Ultra-Low (Can last 10 years on one battery)	High (Drains phone batteries in 1–2 days)



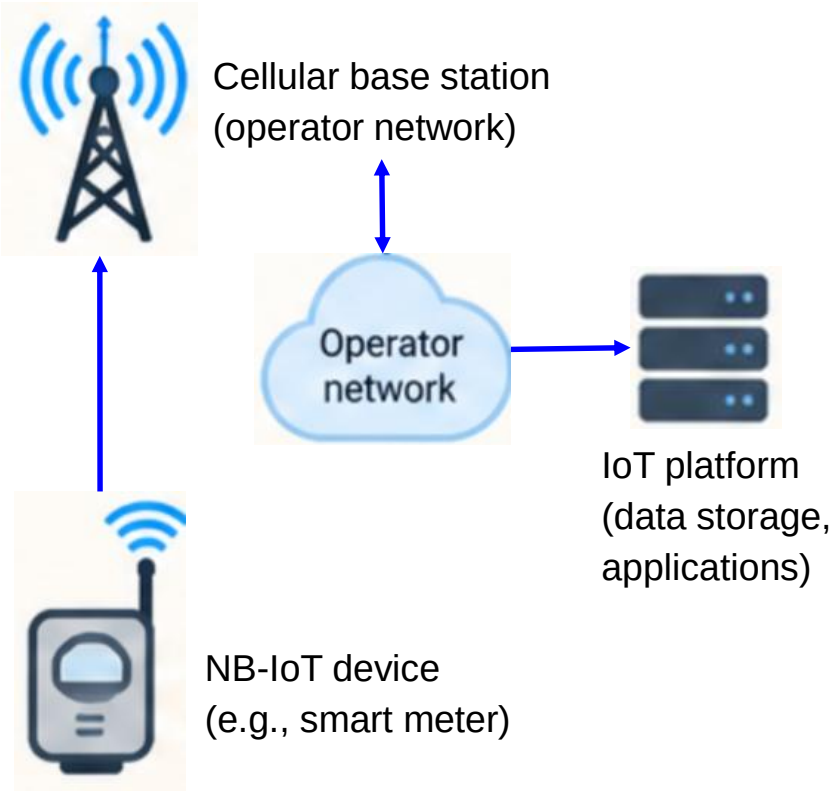
3. Wide-Area Connectivity and Technology Selection

NB-IoT

- **Cellular LPWAN technology**
- Uses **existing cellular network** and operated by **telecom companies**
- Key characteristics:
  - Wide-area coverage (cellular network)
  - Low data rate
  - Low power
  - Good indoor coverage
  - Suitable for relatively static, low-data devices

Typical applications:

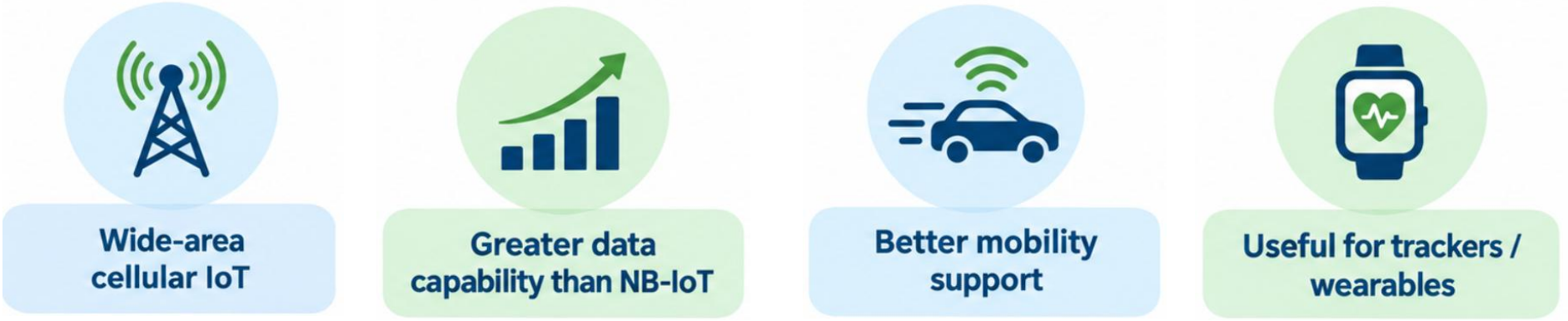




3. Wide-Area Connectivity and Technology Selection

LTE-M

- **High-level cellular IoT technology for connected, mobile things**
- **LTE-M** – machines (sensors, trackers); **LTE** – smartphones, tablets



Application examples



3. Wide-Area Connectivity and Technology Selection

NB-IoT vs. LTE-M

	NB-IoT	**LTE-M**	**LoRa**
Data	lower	higher	very low
Mobility	limited / lower focus	better	limited
Power	low	low–moderate	low
Infrastructure	operator network	operator network	private / local gateway possible
Typical application	meters	trackers	farm / environmental sensors

3. Wide-Area Connectivity and Technology Selection

Full technology comparison

Technology	Range	Data	Power	Mobility	Dependency
Ethernet	Cable	Very high	powered	none	Cabling
Wi-Fi	Local	High	Medium–high	Moderate	AP
BLE	Short	Low–medium	Very low	Good	Phone/ gateway
802.15.4	Short–medium	Low	Low	Limited	Coordinator/ gateway
LoRa	Long	Very low	Very low	Limited/moderate	Gateway
NB-IoT	Wide	Low	Low	Limited	Cellular operator
LTE-M	Wide	Low–medium	Low–medium	Good	Cellular operator
4G/5G	Wide	High	Higher	Excellent	Cellular operator
Satellite	Very wide	Low–medium	Higher	Good	Satellite service

3. Wide-Area Connectivity and Technology Selection

Connectivity selection framework

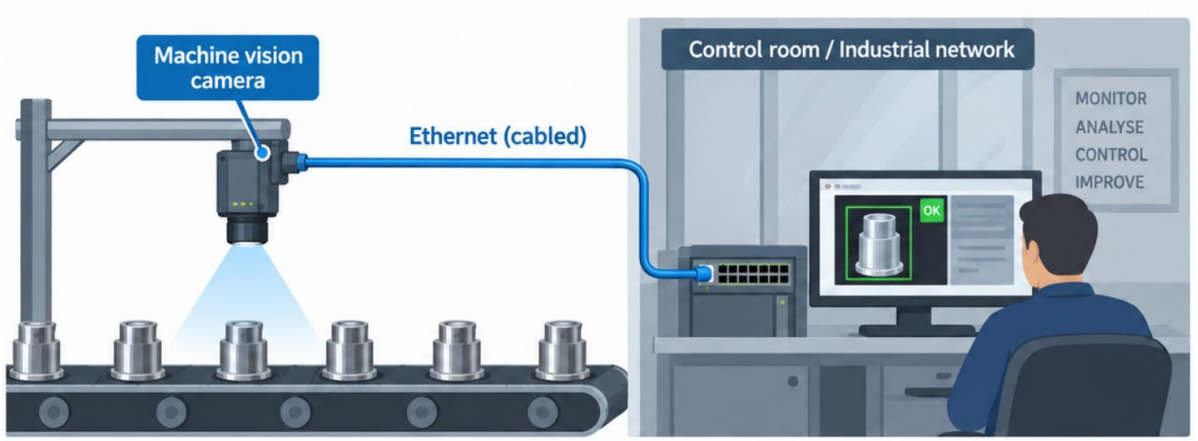
1. How far?
2. How much data?
3. How much power?
4. How quickly?
5. Moving or fixed?
6. How many devices?
7. What infrastructure exists?
8. What environment?
9. What reliability?
10. What cost?

→ **Select technology**

3. Wide-Area Connectivity and Technology Selection

Worked case 1: Machine vision

❖ Industrial camera inspecting a production line, connected to the control room



Requirements

- Fixed installation
- Factory environment
- Video (high data rate)
- Mains power available
- High reliability
- Cable feasible

❖ Reasoning:

- High data – video from camera requires high bandwidth
- Fixed installation
- Power available
- Reliability – Wired Ethernet provides stable connection

Likely candidate:

Ethernet

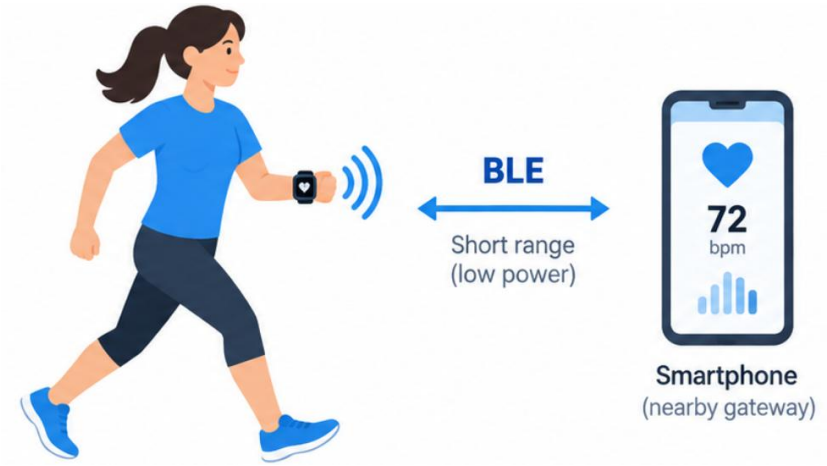
Alternative:

Wi-Fi if cabling is difficult

3. Wide-Area Connectivity and Technology Selection

Worked case 2: Wearable

- Body-worn sensor communicates with smartphone



Requirements

- Body-worn (wearable)
- Battery powered
- Small, periodic data (e.g. heart rate, steps)
- Phone nearby
- Moving user (walking, running, daily activity)

- Reasoning:
  - Short range (meters) is sufficient
  - Small data amounts
  - Low power suits battery operation
  - Smartphone available as nearby gateway

Likely candidate: BLE

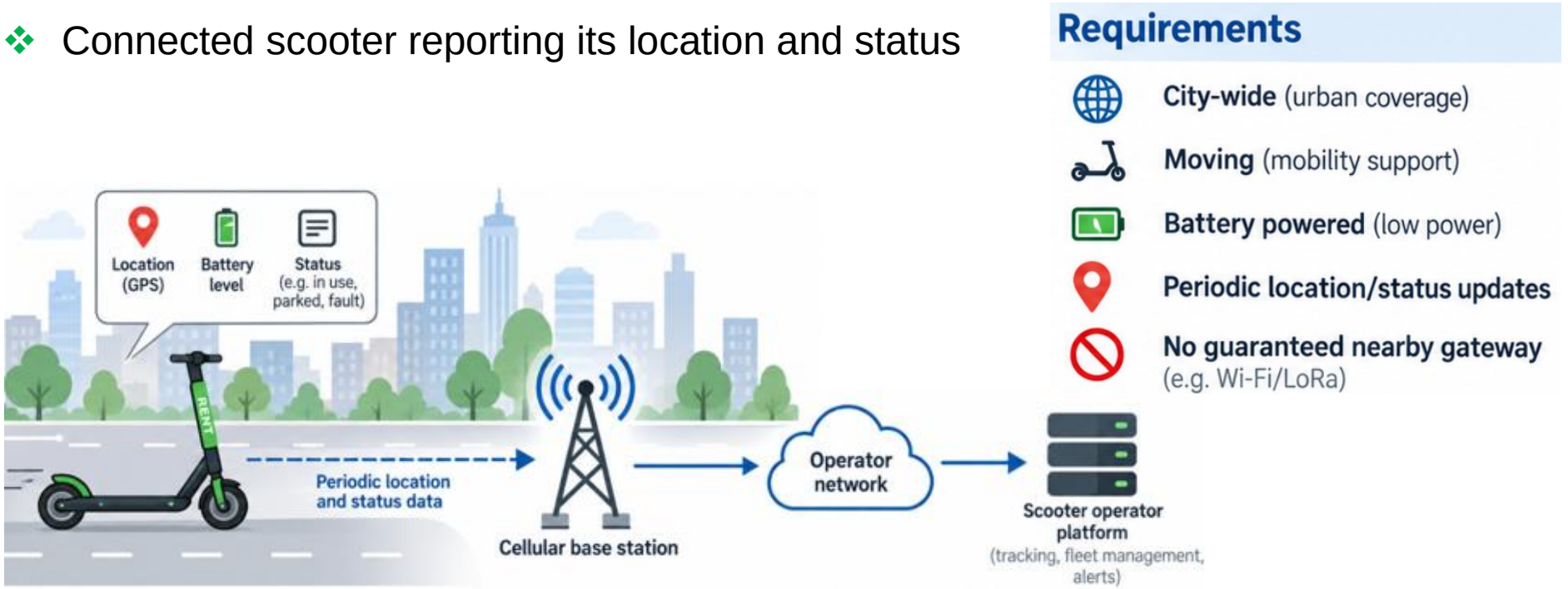
Why not LTE-M?

- Greater complexity (modem, SIM)
- Higher energy and service cost

3. Wide-Area Connectivity and Technology Selection

Worked case 4: Rental scooter

- Connected scooter reporting its location and status



- Reasoning:
  - Wide coverage across city
  - Supports mobility
  - Uses existing operator infrastructure

Likely candidate: LTE-M / cellular

Good balance of coverage, mobility and power efficiency

Summary

Connectivity technology

Requirement

↓

Candidate connectivity technology

↓

Physical / infrastructure fit

↓

Trade-off justification

Device → Connectivity Link → Network / Gateway

Next:

Once the connectivity link exists,

how is information

addressed,

transported

and exchanged?

→ **Communication protocol**